

Sarthak Agrawal

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EXPERIENCE

Software Engineer | VaultWealth (Peak XV) | Feb 2025 – Present

- Created a Financial Planning Service (Go, MySQL) & wrote proprietary logic to calculate clients' financial health score.
- Migrated multiple workflows to Temporal, eliminating 90% of unexpected failures & freeing engineering 3 hrs/day.
- Transitioned webapp from MUI to Tailwind. Built multiple responsive frontend components using React.

Software Engineer | Front.Page (YC S'21) | Jan 2022 – Jan 2025

- Developed real-time pipeline for stock data using Go, Kafka, ProtoBuffers, helped DAU going from 15k to 200k in 14 weeks.
- Built hot news with read/unread tracking, last visit tracking, tripled avg. news scroll count in 3 weeks. Used Clickhouse as data aggregator with Rudderstack as the pipeline.
- Implemented O(1) space/time complexity for retrieval of user's unread news & global notifications count using Redis.
- Improved SEO performance by reducing HTML build & load time from 600ms to 60ms by caching using Redis.
- Implemented queue-based batching, improving database efficiency and reducing load. Optimized user session handling with granular controls, reducing session refresh DB calls by 92%.
- Planned & integrated real-time stock ticks using Socket.io with 3 modes: a) room-based architecture, b) in-memory or c) Redis for user subscription management. Connected with the web via window events.
- Incorporated FCM topic-based notification, reducing delivery time by 90% & increasing delivery rate by 30%.
- Integrated vector-powered personalized feeds using BERT Embeddings, GPT calls, Milvus as vector DB, resulting in 40% more engagement in home feed. Created a pipeline with real-time events from BigQuery for richer user vector.
- Built multiple (support, learning, assistant) RAG based chatbots using OpenAI APIs. Moderation bot ended up reducing human intervention in support queries by 90%.
- Wrote microservice in Node.js with MySQL(Prisma) for complete stock fundamental data lifecycle management.
- Integrated the Razorpay payment gateway with Node.js, resulting in new revenue source & 50% increase in overall revenue.

SKILLS

- **Languages:** Golang, JavaScript/TypeScript, Python, C/C++
- **Databases & Data Tools:** Kafka, Redis, MySQL, PostgreSQL, Elasticsearch, Clickhouse, Milvus, Bigquery
- **Cloud & DevOps:** AWS, GCP, Docker, Kubernetes, Prometheus, Temporal, GenAI

PROJECTS

Stumble - Social App | Node.js, PostgreSQL (with PostGIS), Redis, Docker, AWS, Flutter, Astro, Python

- Launched a platform to help users connect with people nearby, encouraging meaningful in-person interactions.
- App supported real-time chat, location-based profile filtering, face recognition, google auth, push notifications

SignificantHobbies - Life Management App | Next.js, Golang, PostgreSQL, Docker, AWS, Tailwind

- Launched a personal productivity app to manage tasks, habits, food logs, journals, and schedules.
- Designed custom form validation, schedule generation based on user preferences & efficient CRUD operations.

PlayWordle - CLI Puzzle Solver | Golang

- Devised a CLI application to solve New York Times Wordle puzzles with over 97% success rate.

EDUCATION

Manipal Institute of Technology | Aug 2018 – July 2022

B.Tech in Computer Science Engineering

- Relevant Coursework: Algorithms, Data Structures, DBMS, OOP, Operating Systems, Computer Networks
- Projects: Multiple internal portals including Letter of Recommendation, Placement, College Fest & Library Management.

- Technologies Used: HTML, React.js, Next.js, SCSS, Chakra UI, Node.js, GraphQL, Prisma, MongoDB, PostgreSQL, AWS S3